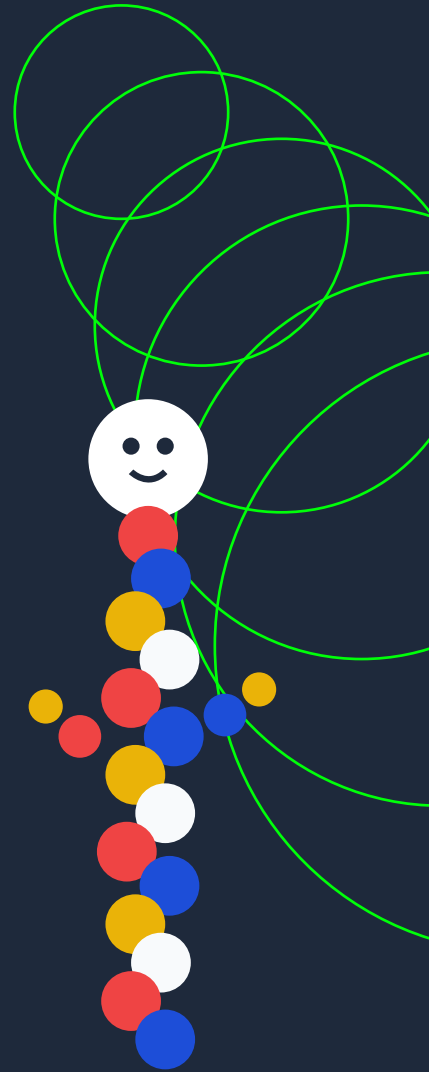


EDUCATE

The Educator Slides Feature

Interactive Topic Slidepacks
with Animated Character

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1 Overview

What We Are Building

An in-app interactive slide system — topic-specific slidepacks for every subject, board, and topic, with The Educator (an animated bead-chain character) floating alongside each slide and narrating the content. Accessible from a dedicated Slides button on the quiz page.

Key Goals

- Topic-specific slidepacks — one per subtopic, not one per subject
- All 26 GCSE subjects across all 4 exam boards: AQA, Edexcel, OCR, WJEC
- The Educator character walks through each slide with narration
- Integrated directly into the existing quiz page flow
- Board-filtered content — AQA slides differ from Edexcel where specs differ
- Works on desktop (side-by-side) and mobile (stacked full-screen)

2 The Educator Character

Visual Design

The Educator is built entirely in SVG — no image files. He is a friendly figure made of 14 coloured beads arranged in a curved helix spine, with a large white circle head, simple cartoon eyes, and a smile arc.

Colour Palette

- Red #ef4444
- Blue #1d4ed8
- Yellow #eab308
- White #f8fafc

Animation States

Idle: Gentle float up/down — CSS keyframes, continuous loop

Talking: Mouth pulses open/closed, slight head tilt, torso beads pulse

Pointing: Right arm beads extend outward toward the slide content

Celebrate: Full body bounce and spin triggered on slide completion

Walking: Slides across screen, beads ripple with 40ms staggered delays



3 Data Architecture

Slide Interface

Field	Type	Description
id	string	Unique identifier
title	string	Slide heading
bullets	string[]	3–5 bullet points per slide
diagram	string (optional)	Mermaid code or SVG string
imageUrl	string (optional)	URL for diagram or photo
narration	string	What The Educator says on this slide
speakerNote	string (optional)	Teacher-facing note

Slidepack Interface

Field	Type	Description
subject	string	e.g. Biology
board	string	AQA Edexcel OCR WJEC
topic	string	e.g. Cell Biology
subtopic	string	e.g. Cell Structure and Microscopy
slides	Slide[]	6–10 slides per subtopic
color	string	Subject accent colour (hex)
estimatedMinutes	number	Estimated time to complete

File Structure

```
src/data/slidepacks/  
  biology/  
    aqa/  
      cell-biology--cell-structure.ts  
      cell-biology--diffusion-osmosis.ts  
    edexcel/ ocr/ wjec/  
  chemistry/ physics/ mathematics/ ...(all 26 subjects)  
  index.ts !• getSlidepacks(subject, board, topic, subtopic)
```

4 Slide Content Structure

Per-Slidepack Template (6–10 slides)

Slide 1: Title slide with learning objectives

Slides 2–3: Core concept explanation with diagrams

Slides 4–5: Key definitions and worked examples

Slides 6–7: Required practicals or calculations (where applicable)

Slide 8: Common exam mistakes students make

Slide 9: Exam-style practice question

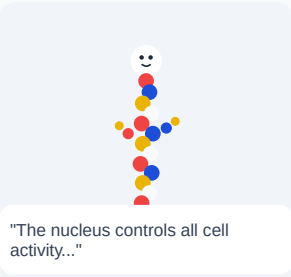
Slide 10: Key points recap and summary

Volume Estimate

- Sciences (Bio/Chem/Phys): 15 topics x 4 subtopics x 4 boards = 240 slidepacks x 8 slides = ~1,920 slides
- All 26 subjects x avg 12 subtopics x 4 boards = ~1,248 slidepacks total
- Estimated total slides across full platform: ~10,000 slides
- Generation: AI batch-generated from spec bullet points, then human-reviewed before going live

Cell Structure & Microscopy

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Cell Structure

- Animal cells: nucleus, membrane, cytoplasm
- Plant cells also have cell wall, chloroplasts
- Bacteria are prokaryotic — no nucleus

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Next %▶

UI Elements

- Subject colour-coded header with topic and subtopic name
- Slide counter (e.g. 2 of 8) and progress bar
- Previous and Next navigation buttons (keyboard: !• !')
- Auto-advance timer option (5 seconds per slide, configurable)
- Mute button to silence voice narration
- Fullscreen toggle and close button (keyboard: Escape)
- Mobile: stacked layout — slide above, The Educator avatar below

Text Bubble (Default — Free)

- Narration text in speech bubble beside The Educator
- Auto-advances on configurable timer
- No API cost — available to all users including free tier

AI Voice (Premium)

- Text passed to OpenAI TTS or ElevenLabs API
- Audio plays while talking animation runs
- Mouth animation syncs to audio playback

Voice Options Comparison

Option	Quality	Cost
Browser speechSynthesis	Basic / Robotic	Free — built in
OpenAI TTS	High quality, 6 voices	~0.001p per slide
ElevenLabs Voice Clone	Most natural, cloned voice	From \$5/month

Quiz Page Integration

A "Slides" button appears in the quiz page header alongside the existing Change Topics button. Clicking it opens the SlideViewer modal pre-loaded with context from the active quiz session.

Context passed to SlideViewer

- Subject — from the URL (e.g. Biology)
- Board — from the URL (e.g. AQA)
- Topic — from sessionStorage (e.g. Cell Biology)
- Subtopic — from sessionStorage (e.g. Cell Structure and Microscopy)

If no slidepack exists for that subtopic, The Educator shows a Coming Soon state. Students can view slides mid-quiz as a revision aid, or review them after getting an answer wrong.

Board Filtering

Subjects requiring board-specific slides

- English Literature — different set texts per board
- History — different depth studies and periods per board
- Geography — different required named case studies per board
- Religious Studies — different faith combinations per board

Subjects shared across boards (minor label differences only)

- Biology, Chemistry, Physics
- Mathematics
- Modern Languages (grammar core is identical)

Phase Plan

1

Foundation Week 1

- Create slides.ts types and slidepack index helper
- Build TheEducator.tsx SVG component (all animation states)
- Build SlideViewer.tsx modal and SlideCard.tsx renderer
- Add Slides button to quiz page header

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The Educator Component Week 1–2

- SVG body: 14 named circle elements in helix pattern
- CSS keyframes: idle, talking, pointing, celebrate, walking
- Staggered animation-delay per bead for wave effect
- State prop with smooth transitions, tested all screen sizes

3

Slide Viewer UI Week 2

- Keyboard navigation (arrows, Escape)
- Responsive: side-by-side desktop, stacked mobile
- Mermaid diagram rendering inside slides
- Auto-advance timer with pause/resume

4

Content Generation Week 2–4

- Priority 1: Combined Science (48 slidepacks)
- Priority 2: Biology, Chemistry, Physics
- Priority 3: Mathematics
- Remaining 23 subjects in batches

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Narration System Week 3

- Text bubble narration (default, free, no API)
- OpenAI TTS API route /api/narrate
- Mouth animation sync to audio.currentTime
- Mute toggle persisted to localStorage

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Board Filtering & Polish Week 4

- Board-aware slidepack lookup
- Coming Soon state for missing slidepacks
- Smooth modal animations, ARIA labels
- Reduced motion support (prefers-reduced-motion)

1

Key Risks & Mitigations

Risk: Massive content volume (26 subjects x 4 boards)

Guard: Generate Sciences first, ship Phase 1, expand iteratively. Never block launch on completeness.

Risk: AI-generated slides contain factual inaccuracies

Guard: Human review pass required before going live. Unreviewed slides flagged with a warning badge.

Risk: TTS voice sounds too robotic for daily use

Guard: Text bubble is always the fallback. Voice is opt-in. ElevenLabs clone for premium users.

Risk: Slidepack not yet available for a topic

Guard: Graceful Coming Soon state. The Educator appears and says slides are being added soon.

Risk: Mobile layout too cramped for side-by-side

Guard: Stacked layout under 768px. The Educator becomes a small floating avatar rather than a full panel.

Risk: Students skip slides and go straight to quiz

Guard: Optional prompt after wrong answer — The Educator asks: "Want me to explain this topic?"

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Verification Checklist

- 01 The Educator renders all 5 animation states correctly on desktop and mobile
- 02 Slides load for Biology / AQA / Cell Biology / Cell Structure with correct narration
- 03 Slides button opens the correct subtopic slidepack based on current quiz context
- 04 Board filter shows AQA vs Edexcel content correctly where specs differ
- 05 Text bubble narration auto-advances through all slides on the timer
- 06 TTS narration plays in sync with The Educator talking animation
- 07 Keyboard navigation works: arrow keys for prev/next, Escape to close
- 08 Mobile stacked layout renders correctly at 375px and 390px screen widths
- 09 Coming Soon state displays correctly for subtopics with no slides yet
- 10 Closing and reopening the viewer remembers the last slide position in session

Ready to build?

The Educator feature will make Educate the only GCSE revision platform with a named, animated mascot that personally guides students through every topic.